

Probability Review

POLS 602

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Why Probability?

- When studying random events from finite samples, we need to express uncertainty about our findings.
- Probability theory is how we can express uncertainty
- Importantly, none of this matters if you don't get identification right!
 - Probability expresses uncertainty about random error. Bias is *nonrandom*
 - Large sample sizes and confidence intervals do not solve the problem of bias

Probability

Two definitions of probability

Frequentist: Proportions of specific events over infinite identical trials.

Bayesian: Subjective beliefs about the relative likelihood of events.

Frequentists assume nothing about the data, Bayesians try to incorporate prior knowledge into their estimates.

Key Terms

Trial: Actions that produce outcomes of interest. Rolling a die

Outcome: The result of a trial. The number rolled on a die.

Event: A set of outcomes. Rolling a 3 is an event with one outcome. Rolling a number less than 3 is an event with two possible outcomes (1 or 2).

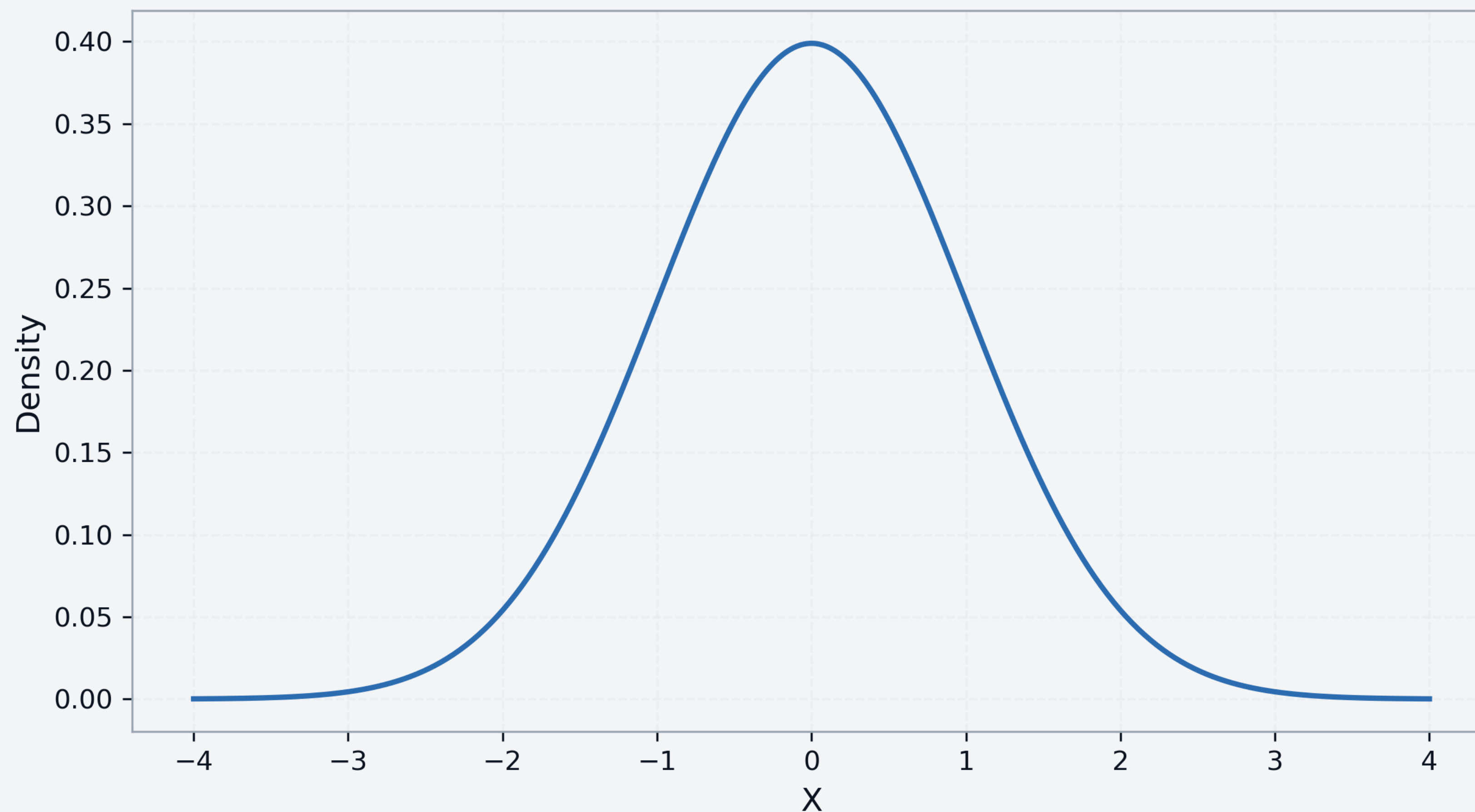
Mutually exclusive events: Events that do not share any outcomes. Rolling a 3 and rolling a number less than 3 are mutually exclusive events.

Sample space: The set of all possible outcomes produced by a trial. Usually denoted Ω . e.g. $\{1, 2, 3, 4, 5, 6\}$

Random Variable: A number assigned to mutually exclusive events produced by a trial.

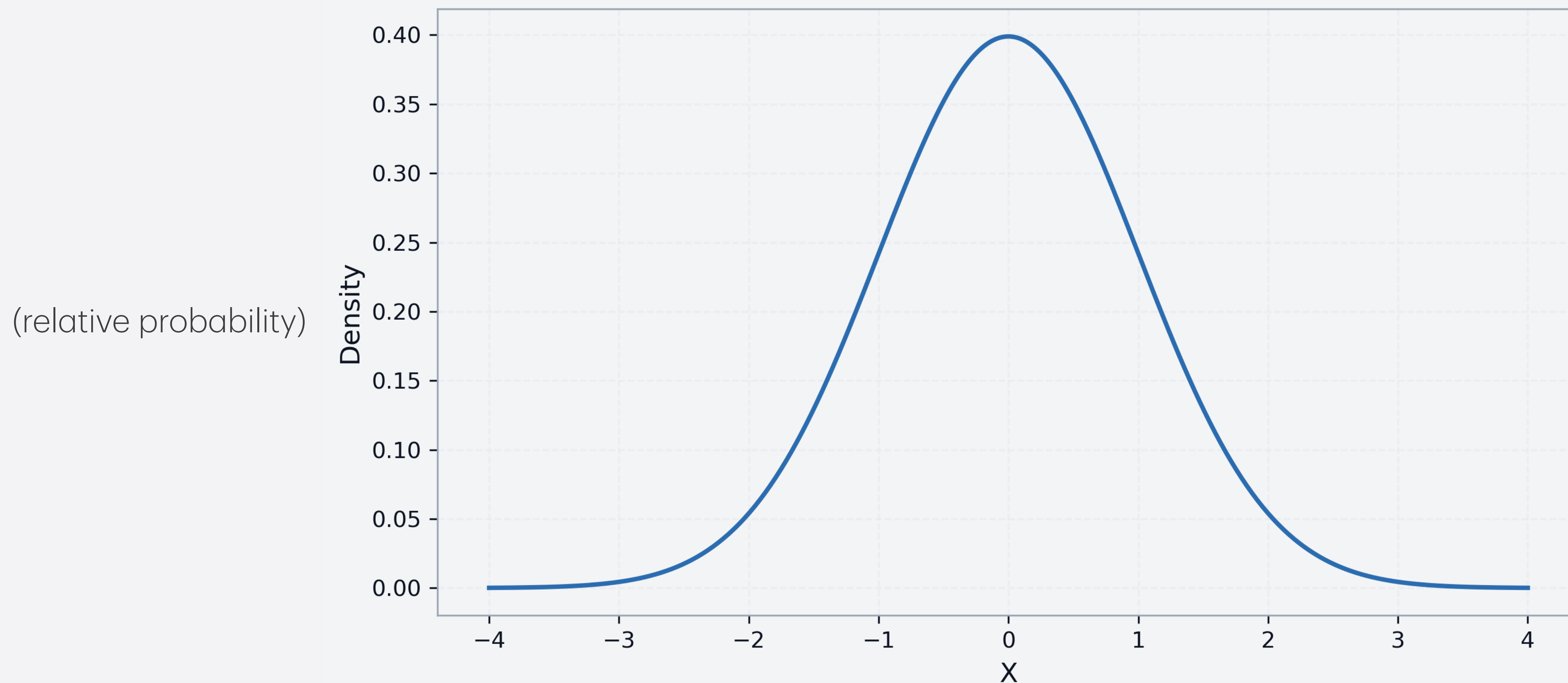
Probability distribution

A mathematical expression of the likelihood of each possible value a random variable can take. The sum of all probabilities in a distribution is 1.

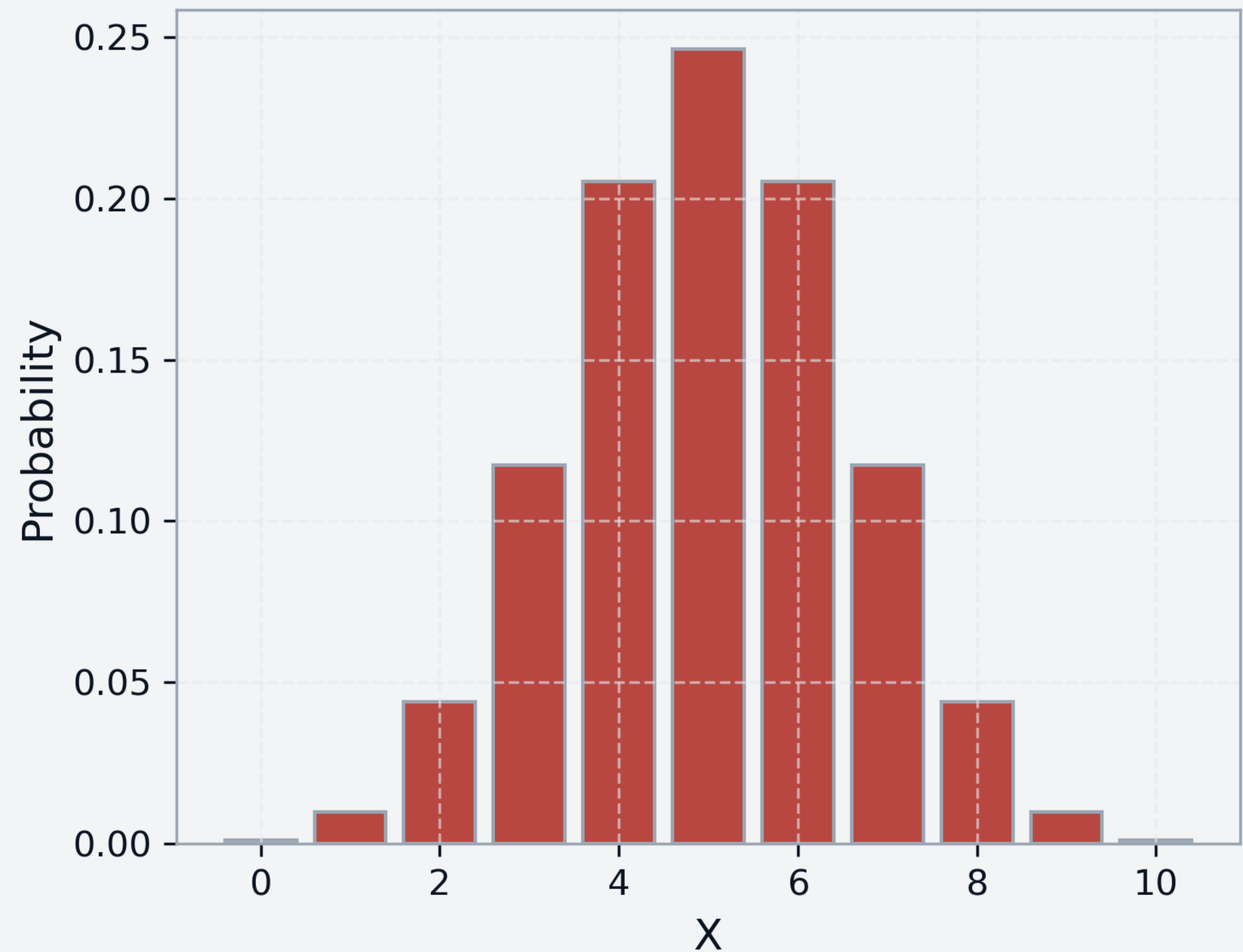
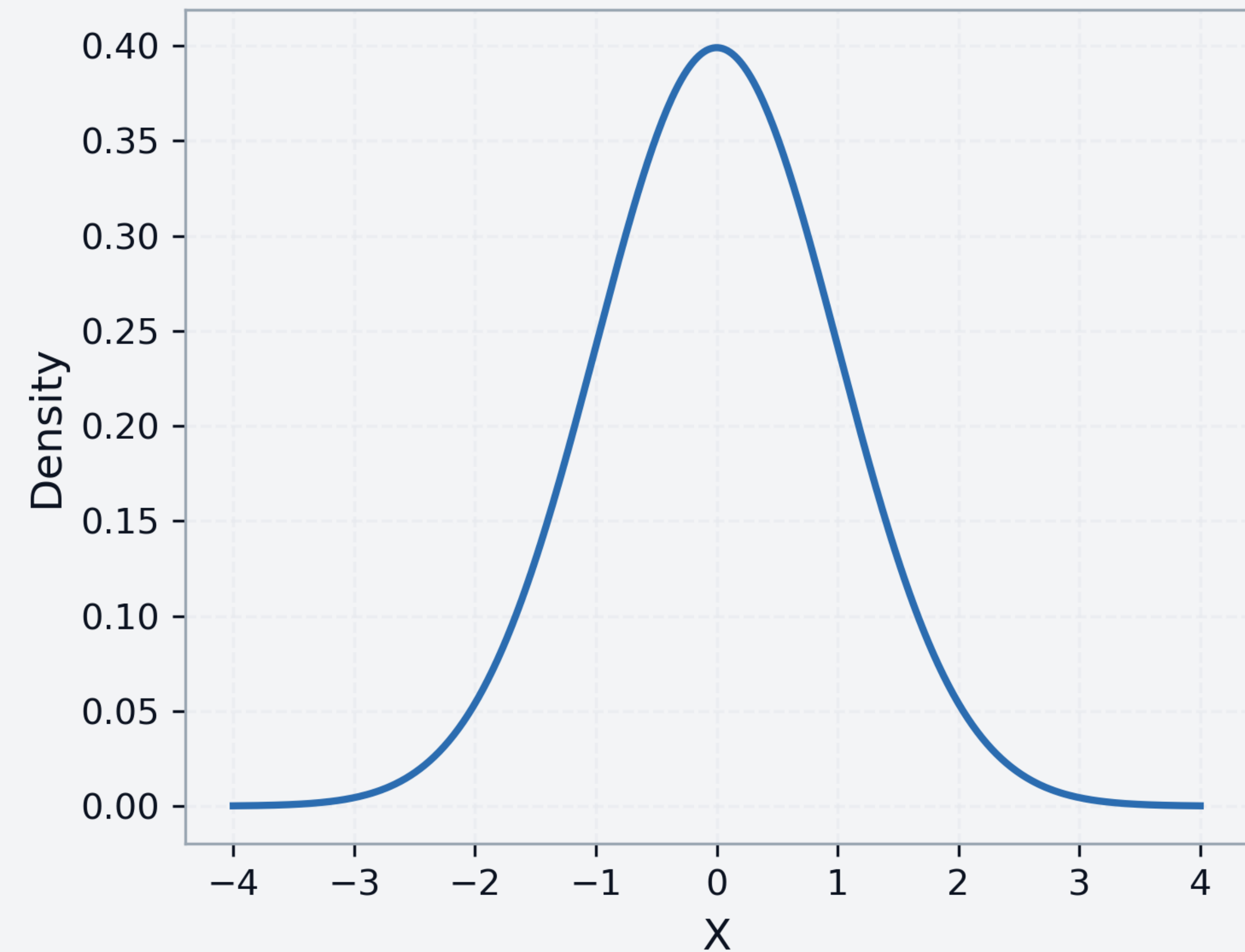


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Probability distribution



Probability density (mass) function

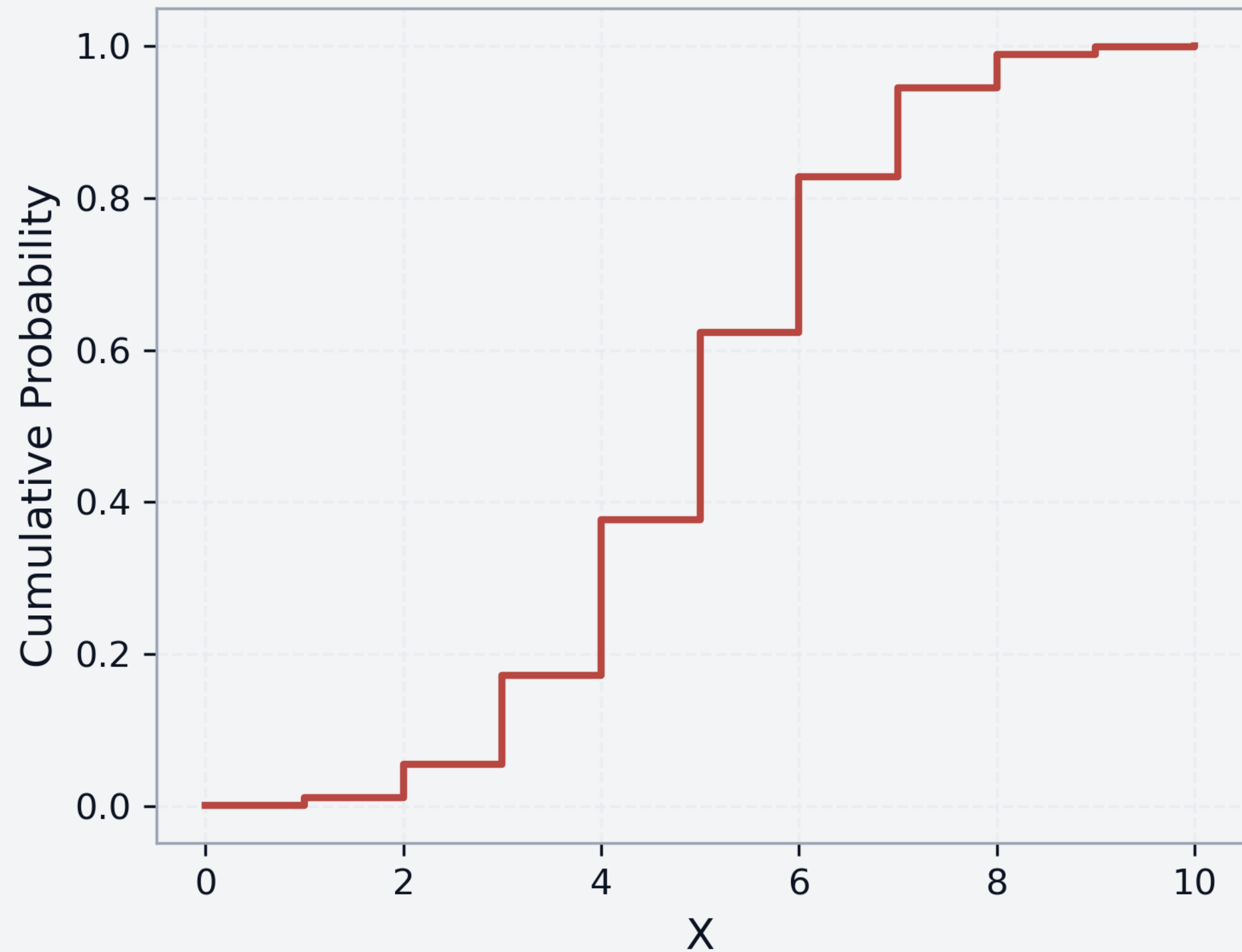
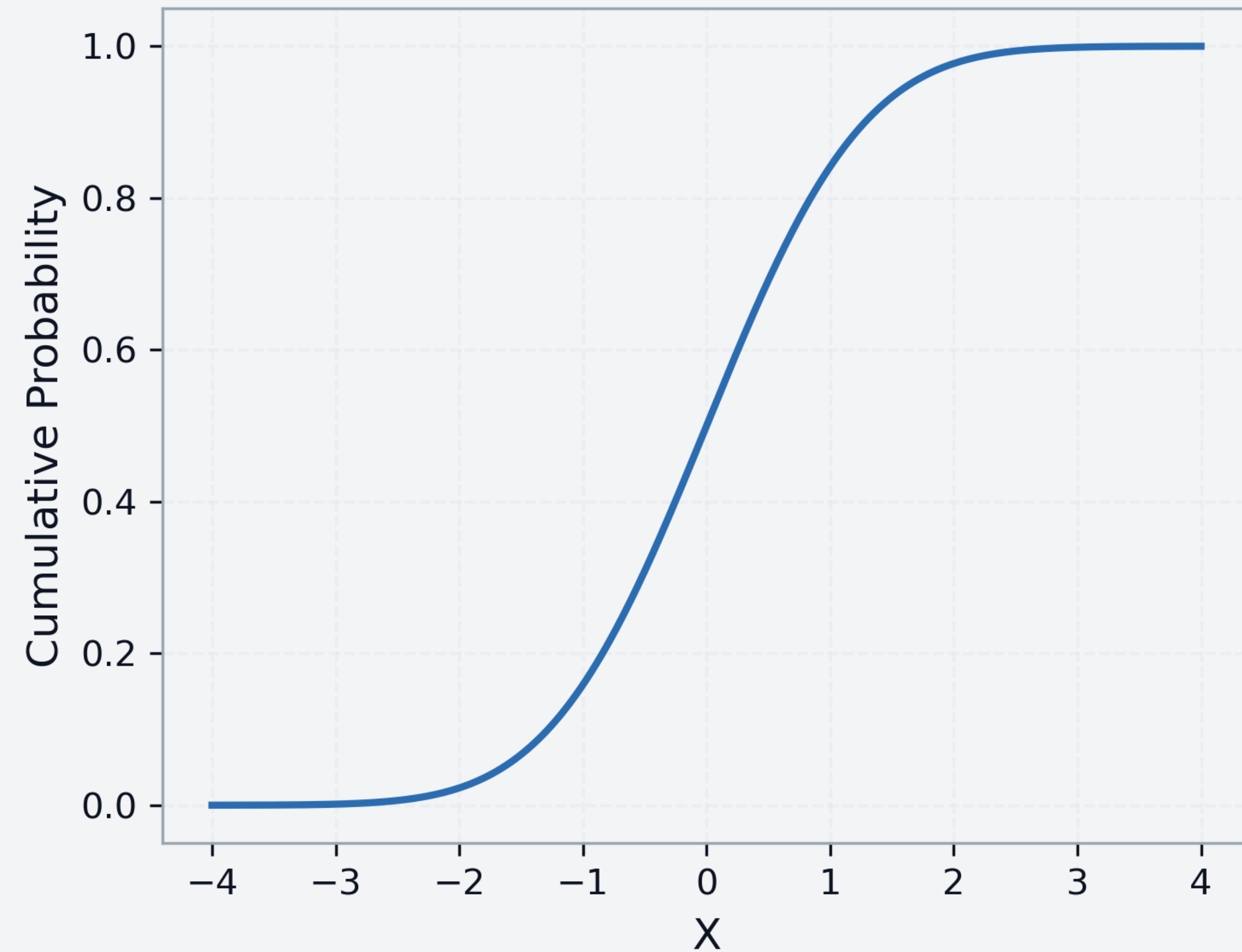
PDF: The mathematical function that provides the height of the density curve for each value of a random variable.

$$f(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) \text{ Normal Distribution}$$

PMF: The mathematical function that provides the probability that a discrete random variable is equal to some value.

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, 2, \dots, n \text{ Binomial Distribution}$$

Cumulative density (mass) function



Probability distributions

Probability distributions are defined by their **parameters**. Parameters determine a distribution's location and spread.

Normal distribution: $N(\mu, \sigma^2)$

Standard normal distribution: $Z(\mu = 0, \sigma^2 = 1)$

Binomial distribution: $B(n, p)$

Bernoulli distribution: $B(n = 1, p)$

Uniform distribution: $U[a, b]$

Poisson distribution: $Pois(\lambda)$

Population Parameters and Sample Statistics

Population Parameters

$$\text{Mean} = \mathbb{E}(X)$$

$$\text{Variance} = \mathbb{V}(X)$$

Sample Statistics

$$\text{Mean} = \bar{X}$$

$$\text{Variance} = \textit{var}(X)$$

The law of large numbers

Given a sample of independent and identically distributed (i.i.d.) values, the sample mean converges to the population mean.

$$\lim_{n \rightarrow \infty} \bar{X}_n = \mathbb{E}(X)$$

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For a binary random variable:

$$\lim_{n \rightarrow \infty} \bar{X}_n = p$$

The central limit theorem

As the sample size increases, the standardized sample mean of X can be approximated by the standard normal distribution.

$$\frac{\bar{X} - \mathbb{E}(X)}{\sqrt{\mathbb{V}(X)/n}} \sim N(0,1)$$

Estimators and Sampling Distributions

Parameter: A true, fixed characteristic of a population. Often a thing we want to estimate. E.g. μ

Estimand: Specific population quantity we want to estimate. Often a parameter like μ , but may also be a causal effect. E.g. the ATE

Estimate: The statistic we compute with sample data, using an estimator.

Estimator: The rule, function, or formula used to calculate an estimate.

Sampling distribution

The distribution that characterizes the variability of the estimator from one sample to another.

- This distribution is approximately normal and centered at the population-level parameter

Standard error

The estimated standard deviation of the sampling distribution.

$$SEM = \frac{\sigma}{\sqrt{n}}$$

Different estimators have different formulas for the standard error. Some are quite complicated or can't be calculated (e.g. the standard error of the median.)

Bootstrapped standard errors

1. Collect data and define your statistic
2. Resample your data with replacement
3. Calculate the statistic for each sample
4. Calculate the standard deviation of your collected statistics.